

CO3 – AT2 -CASE STUDY

**Course Title: Machine Learning Course
Code: ITA0609**

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1. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using Bayesian Belief Networks and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

1. Objective

- Use **wearable sensors** to monitor **heart rate, blood pressure, body temperature, and SpO₂**. Use a **Bayesian Belief Network (BBN)** to detect **health anomalies**.
- Handle **sensor noise and uncertainty** using probability. Evaluate the system using **accuracy, precision, recall, and false-alarm rate**.

2. Bayesian Belief Network

- The main variables are **Heart Rate (HR), Blood Pressure (BP), Body Temperature (BT), SpO₂, Patient Condition, and Health Anomaly**.
- **Conditional dependencies:** HR → Patient Condition , BP → Patient Condition , BT → Patient Condition , SpO₂ → Patient Condition
- **Patient Condition → Health Anomaly**

3. Bayesian Inference – Numerical Analysis

Bayes' theorem is: $P(A | E) = [P(E | A) \times P(A)] / P(E)$

Where: **A** = Health anomaly , **E** = Sensor evidence

Assume:

$$P(A) = 0.10 \text{ , } P(E | A) = 0.80 \text{ , } P(E) = 0.20$$

Therefore,

$$P(A | E) = [0.80 \times 0.10] / 0.20$$

$$= 0.08 / 0.20$$

$$= \mathbf{0.40}$$

Therefore,

$$\mathbf{P(A | E) = 40\%}$$

Result: After observing the sensor evidence, the probability of a health anomaly increases to **40%**.

4. Probability Learning

Consider **1000 historical patient records**.

Total patients = 1000 ,

Abnormal cases = 150,

Normal cases = 850

Prior probability: $P(A) = \text{Number of abnormal cases} / \text{Total cases}$

$= 150 / 1000 = \mathbf{0.15}$

$= \mathbf{15\%}$

Suppose **120 out of 150 abnormal patients** had low SpO₂.

$P(\text{SpO}_2 \text{ Low} \mid A) = \text{Number of abnormal patients with low SpO}_2 / \text{Total abnormal patients}$

$= 120 / 150 = \mathbf{0.80}$

$= \mathbf{80\%}$

Result: The model learns that **80% of abnormal cases** in this dataset are associated with low SpO₂.

5. Conditional Dependency – Numerical Analysis

- Suppose **200 patients** have abnormal heart rate.
- Among them, **120 patients** have a health anomaly.
- $P(A \mid \text{HR Abnormal}) = \text{Number of anomalous patients with abnormal HR} / \text{Total patients with abnormal HR}$

$= 120 / 200 = \mathbf{0.60}$

$= \mathbf{60\%}$

Result: Abnormal heart rate is associated with a **60% probability of anomaly** in this dataset.

6. Sensor Noise and Uncertainty

- Wearable sensors can produce errors due to **patient movement, poor sensor contact, battery problems, and communication errors**. Suppose a noisy SpO₂ reading is **88%**, while HR, BP, and temperature are normal.
- If **SpO₂ is 88%**, HR is high, and temperature is high at the same time, the combined evidence provides stronger support for an anomaly.
- Thus, Bayesian reasoning helps reduce decisions based only on a **single noisy measurement**.

7. Performance Evaluation

Assume the system is tested on **100 patients**.

True Positive (TP) = 36

True Negative (TN) = 54

False Positive (FP) = 6

False Negative (FN) = 4

Accuracy : $\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$

$$= (36 + 54) / (36 + 54 + 6 + 4) = 90 / 100$$

$$= \mathbf{90\%}$$

Precision : $\text{Precision} = TP / (TP + FP)$

$$= 36 / (36 + 6)$$

$$= 36 / 42$$

$$= \mathbf{85.71\%}$$

Recall : $\text{Recall} = TP / (TP + FN)$

$$= 36 / (36 + 4)$$

$$= 36 / 40$$

$$= \mathbf{90\%}$$

False Alarm Rate : $\text{False Alarm Rate} = FP / (FP + TN)$

$$= 6 / (6 + 54)$$

$$= 6 / 60$$

$$= \mathbf{10\%}$$

Analysis: The system achieves **90% accuracy and 90% recall**, while producing a **10% false-alarm rate**.

8. Real-Time Deployment Challenges

- **Sensor Reliability:** Noisy readings can affect predictions.
- **Processing Delay:** Anomalies must be detected quickly.
- **Missing Data:** Wireless failures may cause incomplete readings.
- **Patient Variation:** Normal vital signs differ between patients.
- **Privacy:** Patient information must be securely stored and transmitted.
- **False Alarms:** Excessive alerts can reduce system reliability.

9. Conclusion

- A **Bayesian Belief Network** combines multiple physiological observations to estimate the probability of a health anomaly. The numerical analysis demonstrates how **Bayesian inference, probability learning, and conditional dependencies** can be applied to healthcare data.
- The example achieves **90% accuracy, 85.71% precision, and 90% recall**, showing the potential of Bayesian methods for real-time health anomaly detection.

2.. A cybersecurity system is developed to detect network intrusions using probabilistic learning methods. Apply Naïve Bayes classifiers, EM algorithm, and Bayesian learning to model normal and abnormal network behavior. Analyze how hidden patterns and incomplete data are handled. Evaluate the system's effectiveness in reducing false alarms and improving detection accuracy in dynamic environments.?.

1. Objective

- Detect **normal and abnormal network traffic** using probabilistic learning.
- Apply **Naïve Bayes, EM Algorithm, and Bayesian Learning**.
- Handle **hidden patterns and incomplete data**.

- Reduce **false alarms** and improve **detection accuracy**.

2. Naïve Bayes Classifier

Naïve Bayes calculates the probability that network traffic belongs to a **Normal** or **Attack** class.

For example, consider **1000 network connections**:

- Normal = 800
- Attack = 200

Therefore : $P(\text{Attack})$

$$= 200 / 1000$$

$$= \mathbf{0.20}$$

$$P(\text{Normal}) = 800 / 1000$$

$$= \mathbf{0.80}$$

Suppose an incoming connection has **high packet rate**.

From training data:

$$P(\text{High Packet Rate} \mid \text{Attack}) = \mathbf{0.75}$$

$$P(\text{High Packet Rate} \mid \text{Normal}) = \mathbf{0.10}$$

Using Bayes' theorem:

$$P(\text{Attack} \mid \text{High Packet Rate})$$

$$= [0.75 \times 0.20] / [(0.75 \times 0.20) + (0.10 \times 0.80)]$$

$$= 0.15 / 0.23$$

$$= \mathbf{0.6522}$$

$$= \mathbf{65.22\%}$$

Therefore, the traffic has a higher probability of being an **attack**.

3. EM Algorithm

The **Expectation-Maximization (EM) Algorithm** is useful when some network data is **incomplete or hidden**.

- **E-Step:** Estimate the probability that incomplete records belong to Normal or Attack classes.
- **M-Step:** Update the model parameters using these estimated probabilities.
- The process repeats until the model becomes stable.

Numerical Example

Suppose an incomplete network record has:

$$P(\text{Attack}) = \mathbf{0.70}$$

$$P(\text{Normal}) = \mathbf{0.30}$$

The EM algorithm can use these probabilities instead of simply deleting the incomplete record.

After updating the model, suppose:

$$P(\text{Attack}) = \mathbf{0.78}$$

$$P(\text{Normal}) = \mathbf{0.22}$$

The updated probability indicates that the record is more likely to represent an **attack pattern**.

4. Bayesian Learning

Bayesian learning updates the probability model when **new network evidence** becomes available.

For example:

Prior probability of attack:

$$P(A) = \mathbf{0.20}$$

New evidence indicates suspicious traffic.

If the evidence strongly supports an attack, the posterior probability increases.

Thus, Bayesian learning helps the intrusion-detection system **adapt to changing network behavior**.

5. Handling Hidden Patterns and Incomplete Data

- **Naïve Bayes** classifies traffic using observed network features.
- **EM** estimates hidden information in incomplete records.
- **Bayesian learning** updates probabilities when new traffic patterns appear.

- These methods help detect **unknown or changing attack behavior**.
- Missing values do not necessarily require removing the complete network record.

6. Performance Evaluation

Suppose the system is tested on **1000 connections**:

- $TP = 180$ $TN = 760$
- $FP = 40$ $FN = 20$

$$\textbf{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

$$= (180 + 760) / 1000$$

$$= \mathbf{94\%}$$

$$\textbf{Precision} = TP / (TP + FP)$$

$$= 180 / (180 + 40)$$

$$= \mathbf{81.82\%}$$

$$\textbf{Recall} = TP / (TP + FN)$$

$$= 180 / (180 + 20)$$

$$= \mathbf{90\%}$$

$$\textbf{False Alarm Rate} = FP / (FP + TN)$$

$$= 40 / (40 + 760)$$

$$= \mathbf{5\%}$$

Result: The system achieves **94% accuracy**, **90% recall**, and **only 5% false alarms** in this example.

7. Challenges in Dynamic Environments

- **Changing attack patterns** can reduce model accuracy.
- **Incomplete data** can affect classification.
- **New attacks** may not appear in the training data.

- High network traffic requires **fast processing**.
- Too many **false alarms** can reduce administrator confidence.
- Continuous **Bayesian updating** and model retraining can improve adaptation.

8. Conclusion

Naïve Bayes classifies network traffic, **EM** handles hidden and incomplete information, and **Bayesian learning** updates the model as network behavior changes.

The numerical evaluation gives **94% accuracy, 90% recall, and 5% false-alarm rate**, showing that probabilistic learning can provide effective intrusion detection in dynamic networks.

3. An agriculture-based system aims to predict crop yield under uncertain environmental conditions such as rainfall, soil quality, and temperature. Using Bayesian inference and probability learning, design a predictive model. Analyze how prior agricultural knowledge and seasonal variations influence predictions. Evaluate the model's reliability and discuss its impact on decision-making for farmers and policy planning.?

1. Objective

- Predict **crop yield** under uncertain conditions. Consider **rainfall, soil quality, and temperature**.
- Use **Bayesian inference** and **probability learning**. Study the effect of **prior agricultural knowledge** and **seasonal variation**.
- Evaluate reliability and support **farmer and policy decisions**.

2. Bayesian Predictive Model

The model considers:

Rainfall → Crop Yield

Soil Quality → Crop Yield

Temperature → Crop Yield

The crop yield can be classified as:

- **High Yield**
- **Medium Yield**
- **Low Yield**

Bayesian inference updates the probability of yield based on environmental observations.

Bayes' theorem:

$$P(Y | E) = [P(E | Y) \times P(Y)] / P(E)$$

Where:

Y = Crop yield condition

E = Environmental evidence

3. Bayesian Numerical Example

- Suppose historical agricultural data shows:

$$P(\text{High Yield}) = \mathbf{0.40}$$

- For a season with suitable rainfall, good soil, and suitable temperature:

$$P(E | \text{High Yield}) = \mathbf{0.80}$$

Assume: $P(E) = \mathbf{0.50}$

Therefore,

$$P(\text{High Yield} | E) = [0.80 \times 0.40] / 0.50$$

$$= 0.32 / 0.50 = \mathbf{0.64}$$

$$= \mathbf{64\%}$$

Result: Based on the current environmental conditions, the predicted probability of **high crop yield** is **64%**.

4. Probability Learning

Assume historical records contain **1000 farming seasons**.

- High yield = 400

- Medium yield = 350
- Low yield = 250

Therefore,

$$P(\text{High Yield}) = 400 / 1000$$

$$= \mathbf{0.40}$$

Suppose among the 400 high-yield seasons, **320 had sufficient rainfall**.

$$P(\text{Sufficient Rainfall} \mid \text{High Yield})$$

$$= 320 / 400 = \mathbf{0.80}$$

$$= \mathbf{80\%}$$

This allows the model to learn that sufficient rainfall is strongly associated with high crop yield.

5. Effect of Prior Agricultural Knowledge

Prior agricultural knowledge can improve the initial prediction.

For example:

- Historical records indicate that a particular crop normally has a **40% probability of high yield**.
- This becomes the **prior probability**.
- New rainfall, soil, and temperature observations update this prior.
- Therefore, expert agricultural knowledge and historical data can be combined with current environmental data.

If the prior probability is **40%** and current conditions are favorable, the posterior probability may increase to **64%**, as calculated above.

6. Seasonal Variation

Crop yield changes between seasons because environmental conditions change.

For example:

- **Monsoon season:** Higher rainfall may increase the probability of sufficient water.
- **Summer:** High temperature and low rainfall may reduce yield probability.

- **Winter:** Suitable temperature may improve conditions for certain crops.

Suppose:

- $P(\text{High Yield} \mid \text{Normal Season}) = 0.64$

After observing a drought condition:

- $P(\text{High Yield} \mid \text{Drought}) = 0.30$

The reduction from **64% to 30%** shows how seasonal environmental changes can significantly affect the prediction.

The Bayesian model can update predictions whenever new seasonal information becomes available.

7. Reliability Evaluation

Assume the model is tested on **100 farms**:

- Correct predictions = 88
- Incorrect predictions = 12

Accuracy: = Correct Predictions / Total Predictions

= 88 / 100

= **88%**

Therefore, the model has an **88% prediction accuracy** in this example.

Reliability can also be evaluated using:

- **Prediction Accuracy**
- **Mean Absolute Error (MAE)**
- **Root Mean Square Error (RMSE)**
- **Prediction Confidence**
- **Performance across different seasons**

Lower prediction error and higher accuracy indicate better reliability.

8. Impact on Decision-Making

Farmers

- Helps select suitable **crops and planting periods**.
- Supports **irrigation planning**.
- Helps estimate expected yield before harvesting. Reduces risk caused by uncertain rainfall and temperature. Supports better use of **fertilizers and resources**.

Policy Planning

- Helps estimate **regional crop production**. Supports food-supply planning.
- Helps identify areas at risk of **low agricultural yield**.
- Supports decisions related to **irrigation and agricultural resources**. Can help governments prepare for possible production shortages.

9. Conclusion

- **Bayesian inference** provides a suitable method for crop-yield prediction because it combines prior agricultural knowledge with current environmental evidence. In the example, favorable conditions increased the predicted probability of high yield to **64%**, while drought reduced it to **30%**.
- Therefore, Bayesian probability learning can provide **reliable predictions under uncertainty** and support better decisions for both **farmers and agricultural policy planning**.